

Dissociation of spatial attention and saccade preparation

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Edited by George Sperling, University of California, Irvine, CA, and approved September 20, 2004 (received for review May 19, 2004)

The goal of this experiment was to determine whether the allocation of attention necessarily requires saccade preparation. To dissociate the focus of attention from the endpoint of a saccade, macaque monkeys were trained to perform visual search for a uniquely colored rectangle and shift gaze either toward or opposite this color singleton according to its orientation. A vertical singleton cued a prosaccade, a horizontal singleton, an antisaccade. Saccade preparation was probed by measuring the direction of saccades evoked by intracortical microstimulation of the frontal eye fields at variable times after presentation of the search array. Eye movements evoked on prosaccade trials deviated progressively toward the singleton that was also the endpoint of the correct eye movement. However, eye movements evoked on antisaccade trials never deviated toward the singleton but only progressively toward the location opposite the singleton. This occurred even though previous work showed that on antisaccade trials most neurons in frontal eye fields initially select the singleton while attention is allocated to distinguish its shape. Thus, sensorimotor structures can covertly orient attention without preparing a saccade.

frontal eye fields | microstimulation | target selection

Humans can orient to an interesting object or region of the visual scene covertly by allocating attention and overtly by executing an eye movement. A longstanding issue has been whether the shift of attention and the shift of gaze are independent (1). The premotor theory of attention (2, 3) posits that the allocation of spatial attention is equivalent to planning but not executing a saccade. Evidence for this theory includes the coupling of spatial attention and saccade preparation (4–6), observations that neurons in sensorimotor structures such as frontal eye fields (FEF) are modulated when attention is allocated (7–11), that the trajectories of saccades can be influenced by the allocation of attention (12), and that electrical stimulation of FEF and superior colliculus can influence the allocation of attention (13, 14).

However, other evidence suggests a functional distinction between covert and overt orienting (15). It is possible to shift attention without shifting gaze (5, 6). Also, the selective activity of visually responsive neurons in sensorimotor structures corresponds to the allocation of attention distinct from saccade preparation (16–19). In particular, neurons in FEF select the location of a salient object in an array when monkeys maintain fixation or shift gaze away from that object (7, 20–23). Furthermore, in a stop signal task, visual neurons in FEF and superior colliculus do not produce signals sufficient to contribute to the control of saccade generation (24, 25).

The properties of neurons in the FEF of monkeys performing visual search requiring explicit stimulus–response mapping based on the shape of a color singleton provide an opportunity to investigate whether orienting to a visual stimulus necessarily requires preparing a saccade to that stimulus (20). Monkeys were trained to perform a color singleton visual search task with a prosaccade or antisaccade response cued by the orientation of the singleton (Fig. 1*A*). Single-unit recordings from FEF during this task showed that in prosaccade search trials, most neurons

select the location of the singleton that was also the endpoint of the saccade (Fig. 1*B Left*), typically ≈ 100 ms after the presentation of the array (20). In antisaccade trials, most neurons initially select the singleton (also ≈ 100 ms) then undergo a dramatic modulation to select the endpoint of the saccade (≈ 200 ms; Fig. 1*B Right*). The period of selection of the singleton almost certainly corresponds to the allocation of attention (7).

Converging lines of evidence link visual attention with activity in the FEF. First, visual neurons in FEF signal the location of a singleton under conditions that have been shown to automatically attract attention (26, 27). This selection emerges even when gaze remains fixed or shifts to a location out of the response field (20–23). Second, weak electrical stimulation of macaque FEF has been shown to influence the allocation of attention and bias visual processing in extrastriate cortex in retinotopically matched sites (13). Third, functional imaging studies have shown that human FEF is active as attention is shifted, even when no eye movements are made (10, 28–30). Finally, transcranial magnetic stimulation delivered to the human FEF influences performance of visual search and visual attention tasks (31, 32).

Saccade preparation was probed in this experiment by using intracortical microstimulation. FEF stimulation evokes fixed-vector saccades in the absence of visual stimulation (33), but more recent studies have demonstrated that evoked saccades can be influenced by the preparation of a saccade to perform a task (14, 34, 35). The search array was arranged such that the saccades evoked by microstimulation of FEF in each experimental session were orthogonal to the axis of the stimuli guiding prosaccades or antisaccades. The evolution of saccade preparation was assessed by measuring the deviation of the saccades evoked by microstimulation at different times after search array presentation. In prosaccade trials, the deviations should increase progressively toward the singleton that is also the endpoint of the correct saccade (Fig. 1*C Left*). In antisaccade trials, the evoked saccades should deviate ultimately toward the endpoint of the saccade opposite the singleton. The key hypothesis of this experiment was evaluated by determining whether saccades evoked at intermediate times (when the singleton, but not yet the endpoint, was selected) deviated toward the singleton or the endpoint (Fig. 1*C Right*).

Eye movements evoked by stimulation during the task were quantified by the angular difference from the vector of the saccade evoked at the earliest stimulation time (0–60 ms) used during each session before any neural selection had occurred. Angular deviations toward the singleton were assigned positive values; deviations away from the singleton were assigned negative values (Fig. 1*D*). Accordingly, on prosaccade trials, the

This paper was submitted directly (Track II) to the PNAS office.

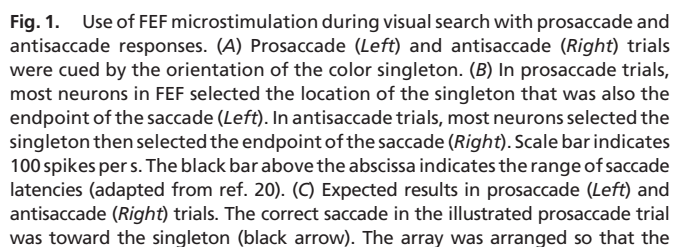
Abbreviation: FEF, frontal eye fields.

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FEF microstimulation with tungsten microelectrodes (FHC, 2–4 M Ω ; 60-ms trains of 0.2-ms biphasic pulses at 500 Hz) was delivered on 50% of randomly interleaved prosaccade and antisaccade trials. Results were the same with lower fractions of

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significantly different from 0°, an ANOVA was performed to define the 95% confidence interval. For prosaccade trials, stimulation at least 120 ms after array presentation evoked saccades with significant deviations. For antisaccade trials, deviations were significant after 140 ms. These values were almost identical for the monkeys' data examined individually [monkey L, 120 ms (pro) and 140 ms (anti); monkey P, 120 ms (pro) and 130 ms (anti)]. The deviations became significant only after the orientation of the singleton was encoded but before the endpoint of the saccade was selected, as assessed by the time of modulation of FEF neurons (7, 20).

It is possible that on individual antisaccade trials, the deviations may have been biased initially toward the singleton, followed by a reversal to the saccade endpoint (i.e., the mean deviations may conceal more subtle deviations that may have occurred on some fraction of trials). To quantify this, the variability in the endpoints of saccades evoked in antisaccade trials was compared to the baseline dispersion of the endpoints of the saccades evoked in prosaccade trials at the earliest stimulation time before any systematic deviation occurred. Deviations in antisaccade trials exceeding the 95th percentile of the distribution of baseline deviations toward the singleton were vanishingly rare (monkey L, 0.032%; monkey P, 0.020%).

Discussion

In the present task, attention is allocated initially to the location of the singleton because it is conspicuous and its shape must be resolved to produce the appropriate response (26, 27). Our results are consistent with the evidence that target selection is not sufficient for saccade preparation. The absence of deviations

toward the singleton when it was being selected by neurons in FEF during attention allocation demonstrates that saccade preparation is not an obligatory or immediate outcome of visual selection and so challenges the premotor theory of attention. Evidence using transcranial magnetic stimulation of the FEF in humans (31) and electrically stimulating the intermediate layers of the macaque superior colliculus (38) reinforces this conclusion. Also, the independence of preparing an eye movement and allocating attention has been demonstrated in dual-task paradigms when shifting attention is volitional (15), as well as when attention gets captured reflexively by a stimulus with an abrupt onset (39). Consequently, the deviations of the endpoints of evoked saccades reveal only the state of saccade preparation and do not necessarily measure the moment-by-moment locus of attention within the visual field (14) or the current state of sensory evidence from which saccade production is derived (34). Covert attention and overt gaze may be linked under many conditions, but the present results demonstrate that the link is not obligatory or immediate (40). Such a dissociation can come about if different pools of neurons within the network of sensorimotor structures convey distinct signals. Identifying such distinctions is necessary to elucidate the proper mapping between cognitive processes and neural processes.

We thank M. Mebane for his assistance in data collection and J. Gold, R. Marois, T. Palmeri, A. Roe, and A. Rossi for constructive comments on the manuscript. This work was supported by Robin and Richard Patton; National Science Foundation Grant BSC0218507; and National Institutes of Health Grants RO1-EY08890, F32-EY14502, T32-EY007135, P30-EY08126, and P30-HD015052.

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